

Program : Diploma in Instrumentation Engineering	
Course Code : 2089	Course Title: Electronic Instrumentation Lab
Semester : 2	Credits: No Credit
Course Category: Engineering Science	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the static characteristics of a given instrument
- To familiarize the use of DC and AC bridges
- To learn the calibration of watt meter and energy meter
- To familiarize the use CRO/DSO and strip chart recorder.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of electronic components		Fundamentals of Electrical & and electronics	Secondary School
Fundamentals of electric and magnetic circuits		Fundamentals of Electrical & and electronics	1
Fundamentals of ac circuits		Fundamentals of Electrical & and electronics	1

Course Outcomes :

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Determine the static characteristics of a given instrument and to convert the given galvanometer in to ammeter and voltmeter.	12	Applying
CO2	Determine unknown resistance, capacitance and inductance values using appropriate bridges.	8	Applying
CO3	Calibrate given watt meter and energy meter.	8	Applying
CO4	Explain the use of CRO/DSO and recorders.	14	Understanding
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						3
CO2	3	3					
CO3	3	3		3			
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	To determine the static characteristics of a given instrument and to convert the given galvanometer in to ammeter and voltmeter.		
M1.01	Calibrate the given instrument (ammeter-using slide wire potentiometer /voltmeter-using slide wire potentiometer /any other instrument) and determine accuracy, error, precision, hysteresis, sensitivity.	6	Applying
M1.02	Convert the given galvanometer to ammeter and voltmeter.	6	Applying
CO2	To determine unknown resistance, capacitance and inductance values using appropriate bridges.		
M2.01	Determine the unknown resistance using Wheatstone's bridge.	2.5	Applying
M2.02	Determine the unknown Inductance using Hays bridge.	2.5	Applying
M2.03	Determine the unknown Capacitance using Schering Bridge	3	Applying
	Lab Exam I	1.5	
CO3	To calibrate given watt meter and energy meter.		
M3.01	Calibrate wattmeter by direct loading.	3	Applying
M3.02	Calibrate energy meter by direct loading at unity power factor.	5	Applying
CO4	To familiarize the use of CRO/DSO and recorders		
M4.01	Utilize the CRO and DSO for voltage, time period and frequency measurement.	6	Applying

M4.02	Measure the unknown frequency by using known frequency by Lissajous pattern method.	3	Applying
M4.03	Measure phase difference of given waveforms using CRO.	2	Applying
M4.04	Illustrate Strip Chart Recorder.	3	Understanding
	Lab Exam II	1.5	

Text / Reference

T/R	Book Title / Author
R1	B L Thereja, “ Electrical Technology”, S Chand.
R2	AK Sawhney, “A Course in Electrical and Electronic Measurements and Instrumentation”, Dhanpath Rai.
R3	Kalsi, “ Electronic Instrumentation”, TMH.
R4	William D Cooper, “ Modern Electronic Instrumentation and Measurement Techniques”, PHI.
R5	David A. Bell, “ Electronic Instrumentation and Measurements”, Second Edition, Oxford University Press.

Online Resources:

Sl.No	Website Link
1	www.ieee-ims.org
2	www.nptel.ac.in/courses
3	http://vlab.co.in/
4	https://www.electronicweekly.com/
5	http://www.instrumentationtoolbox.com/